

Attenuation of AC Magnetic Wave Propagating Through Aluminum Pipe

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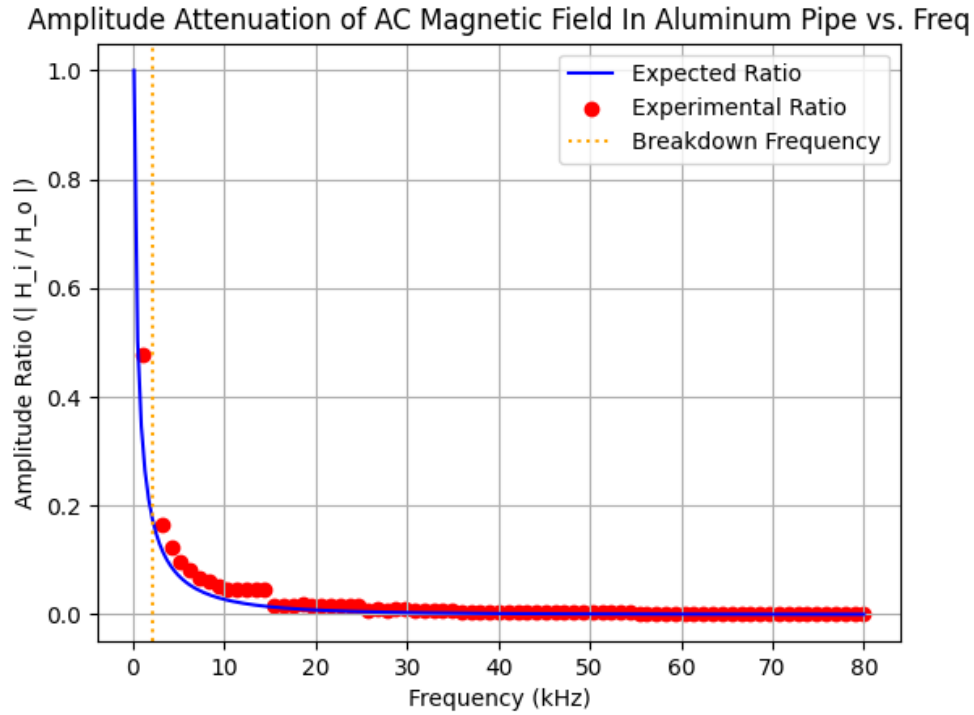


Figure 1: This plot shows the amplitude attenuation of an alternating magnetic wave propagating through an aluminum pipe as a function of frequency, which happens due to the skin effect phenomenon. The solid blue line shows the theoretical amplitude ratio while the red data points show the experimentally measured attenuation. Here H_o denotes the amplitude of the magnetic field on the surface of the pipe and H_i represents the amplitude on the inner boundary. An additional vertical dotted line is shown in yellow to mark the limit of the theoretical model; experiments were collected primarily beyond this breakdown frequency (~ 2.3 kHz) to show the convergence of theory and experimental data at higher frequencies.

1 Related Theory

Skin effect is a phenomenon that occurs in conductive material (eg. aluminum metal), characterized by the attenuation of an external AC magnetic field when it tries to pass through. Since electrons are free to rearrange in conductors to counteract external fields (Lenz's law), eddy currents are produced to generate an opposing magnetic field. This "shielding" effect is typically quantified by the skin depth (δ), the distance measured from the surface of a conductor at which the field amplitude drops to $1/e$ (approximately 37%) of its surface value. As frequency increases, the skin depth decreases, forcing the magnetic field to concentrate near the outer surface and leading to the decay in the internal field (H_i) shown in the Figure 1. The theoretical amplitude ratio of the inner/outer magnetic field is given by:

$$\frac{H_i}{H_o} = \rho e^{i\phi}$$

where

$$\begin{aligned}\phi &= k_o(R_2 - R_1) + \arctan\left(\frac{R_1 k_o}{2 + R_1 k_o}\right) \\ \rho &= 2\sqrt{\frac{R_2}{R_1}} \left\{ \frac{e^{-k_o(R_2 - R_1)}}{\sqrt{1 + R_1 k_o + \frac{R_1^2 k_o^2}{2}}} \right\} \\ k_o &= \sqrt{\frac{\omega \sigma \mu}{2}}\end{aligned}$$

2 Experimental Procedure

I used a function generator to drive an 8Vpp sinusoidal waveform into a primary copper coil, which acted as the input alternating magnetic field (H_o). I then placed a smaller receiver coil inside the primary coil to detect the magnetic flux. In an attempt to normalize the data, I first took measurements in "free space" to establish the baseline H_i , and then repeated with the aluminum pipe inserted between the primary and receiver coils to measure the internal field H_i . In order to visualize the effects of frequency, I configured the function generator to sweep across 80 equally-spaced input frequencies between 100Hz to 80kHz. With this data, I was able to compute the amplitude ratio $|H_i/H_o|$ as a function of frequency (Figure 1).

3 Results

The biggest standout factor for me from Figure 1 is seeing the rapid, non-linear relationship between the amplitude ratio $|H_i/H_o|$ decaying and the input frequency increasing. This confirms the presence of the skin effect. Another observation is that the experimental amplitude ratio converges well to the theoretical curve at higher frequencies, but struggles at lower frequencies below the frequency breakdown point. This discrepancy arises due to the edge effects and perhaps because the skin depth exceeds the wall thickness at low frequencies, allowing a small portion of the magnetic field to propagate through the material relatively unimpeded and/or from the ends of a finite pipe. If I were to redo this aspect of the experiment, I would increase the sampling density in the lower frequency range to better characterize the behavior in this range all while using a longer aluminum pipe.